

“Calculus 3”

Multi-Variable Calculus

Instructor: Álvaro Lozano-Robledo

Day 12

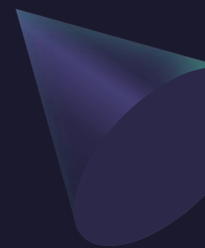


Any Reminders? Any Questions?

- I have regular office hours Mondays via Webex 1-2pm
- I have regular office hours Thursdays in-person 3:30-4:30pm
- Calc 3 Calc Night: MONT 104 at 6:30-8:30pm on Thursdays!



ALVARO: Start the recording!



“Calculus 3”

Multi-Variable Calculus

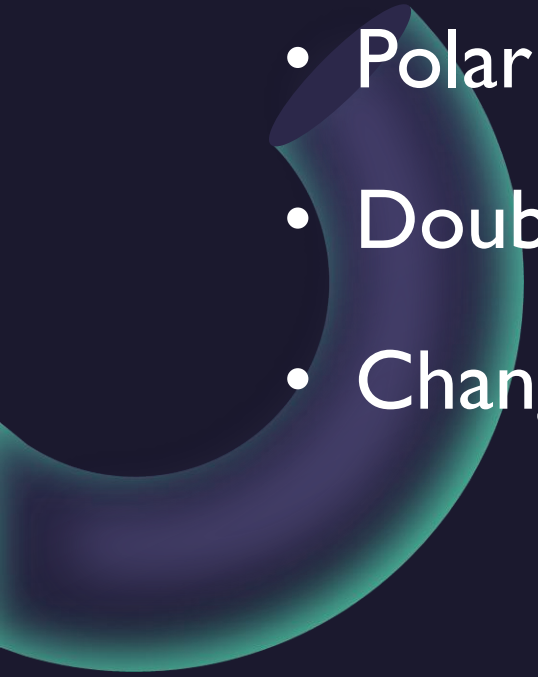
Instructor: Álvaro Lozano-Robledo

Double Integrals in Polar





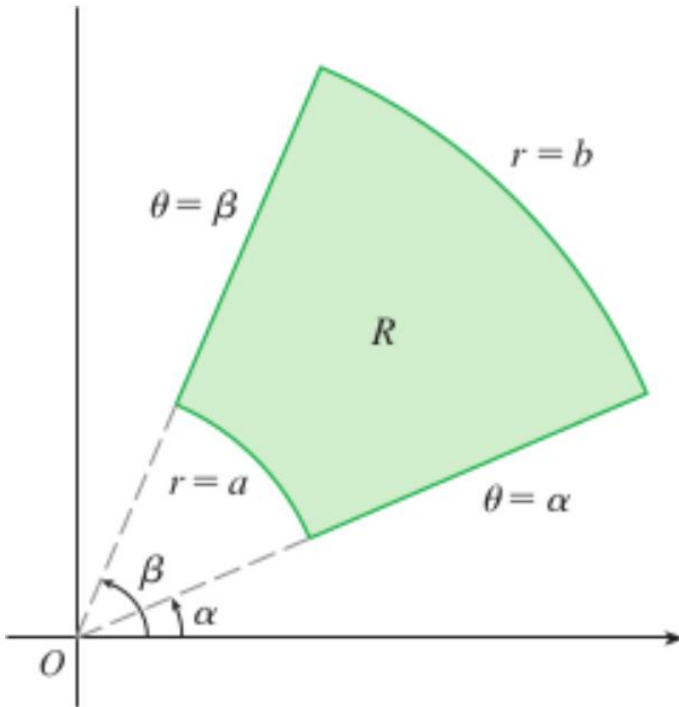
Today – Double Integrals in Polar Coordinates!

- 
- Polar Coordinates
 - Double Integrals in Polar Coordinates
 - Change from Cartesian to Polar

Double Integrals in Polar Coordinates

If f is continuous on a polar rectangle R given by $0 \leq a \leq r \leq b$, $\alpha \leq \theta \leq \beta$, where $0 \leq \beta - \alpha \leq 2\pi$, then

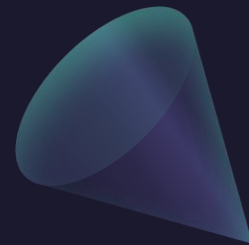
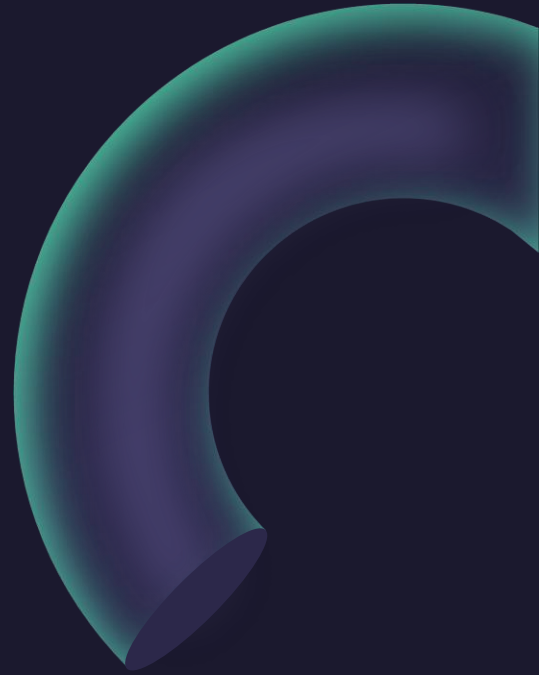
$$\iint_R f(x, y) dA = \int_{\alpha}^{\beta} \int_a^b f(r \cos \theta, r \sin \theta) r dr d\theta$$



Example: Evaluate the value of the double integral

$$\iint_R (x + x^2 + y^2) dA$$

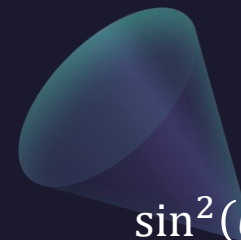
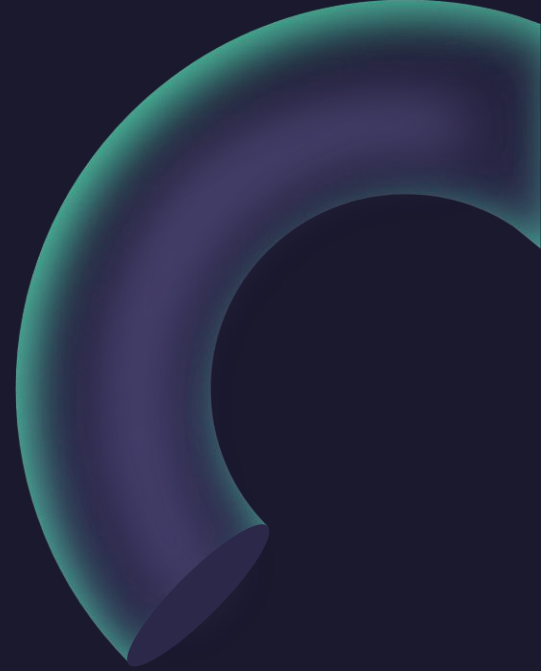
where R is the first quadrant of a circle of radius 2.



Example: Evaluate the value of the double integral

$$\iint_R (x + x^2 + y^2) dA$$

where R is the first quadrant of a circle of radius 2.



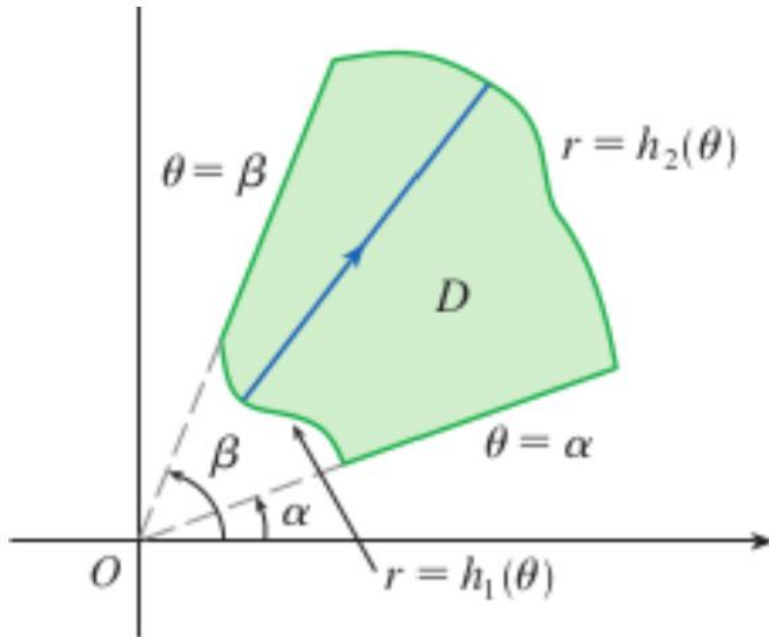
$$\sin^2(\theta) = \frac{1 - \cos(2\theta)}{2}$$

Integrals in Polar Coordinates of Type I

$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$

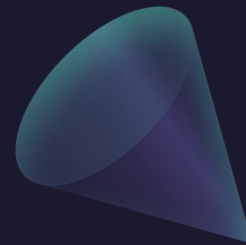
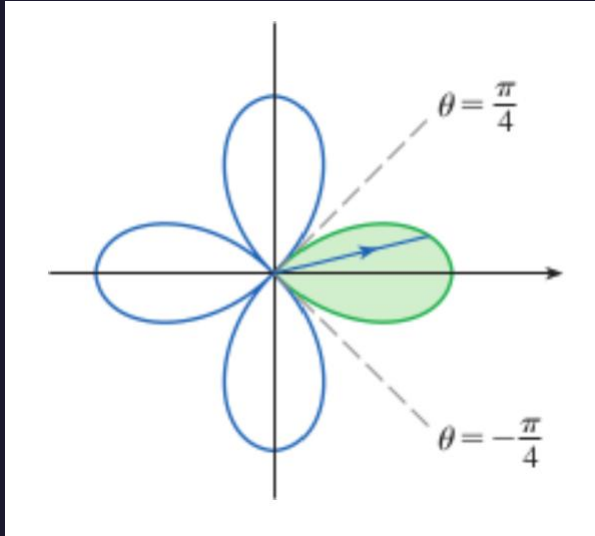
3 If f is continuous on a polar region of the form

$$D = \{(r, \theta) \mid \alpha \leq \theta \leq \beta, h_1(\theta) \leq r \leq h_2(\theta)\}$$



$$\iint_D f(x, y) \, dA = \int_{\alpha}^{\beta} \int_{h_1(\theta)}^{h_2(\theta)} f(r \cos \theta, r \sin \theta) \, r \, dr \, d\theta$$

Example: Find the area enclosed by one loop of the four-leaved rose
 $r = \cos(2\theta)$



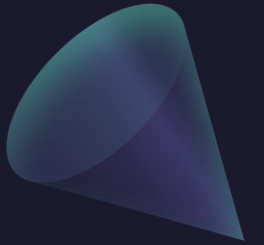
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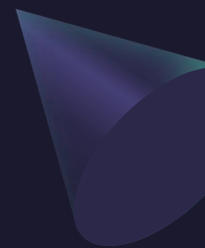
$$\cos^2(\theta) = \frac{1 + \cos(2\theta)}{2}$$

Questions?





ALVARO: Start the recording!



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Multi-Variable Calculus

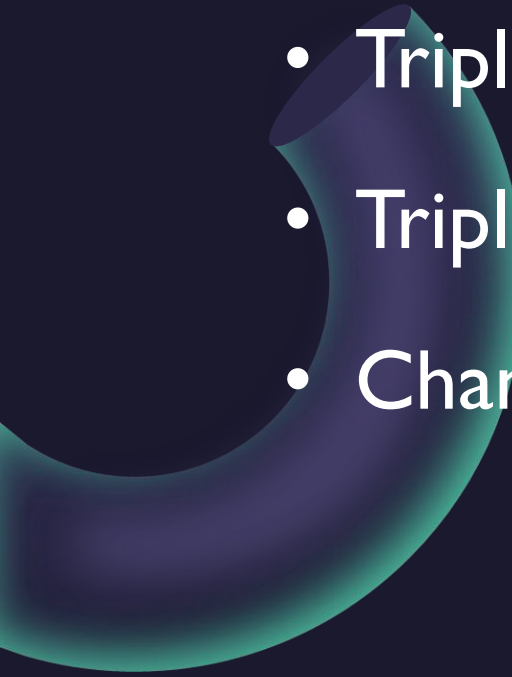
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Triple Integrals

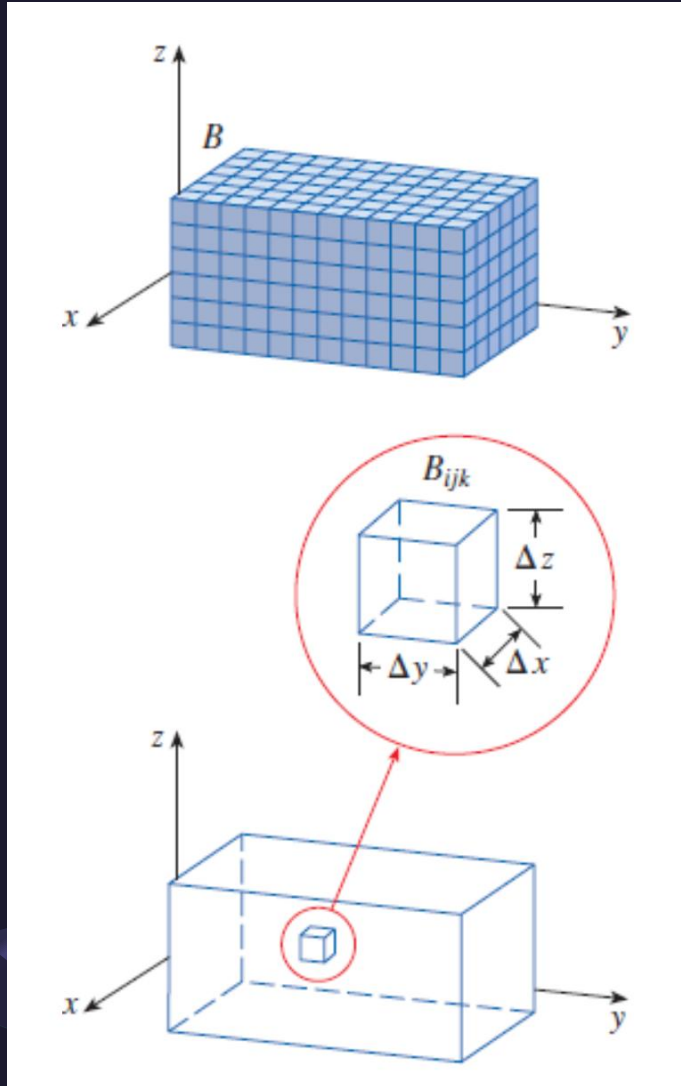




Today – Double Integrals in Polar Coordinates!

- 
- Triple Integrals in Boxes
 - Triple Integrals in Regions
 - Changing the Order of Integration

Riemann Integral in 3 Variables



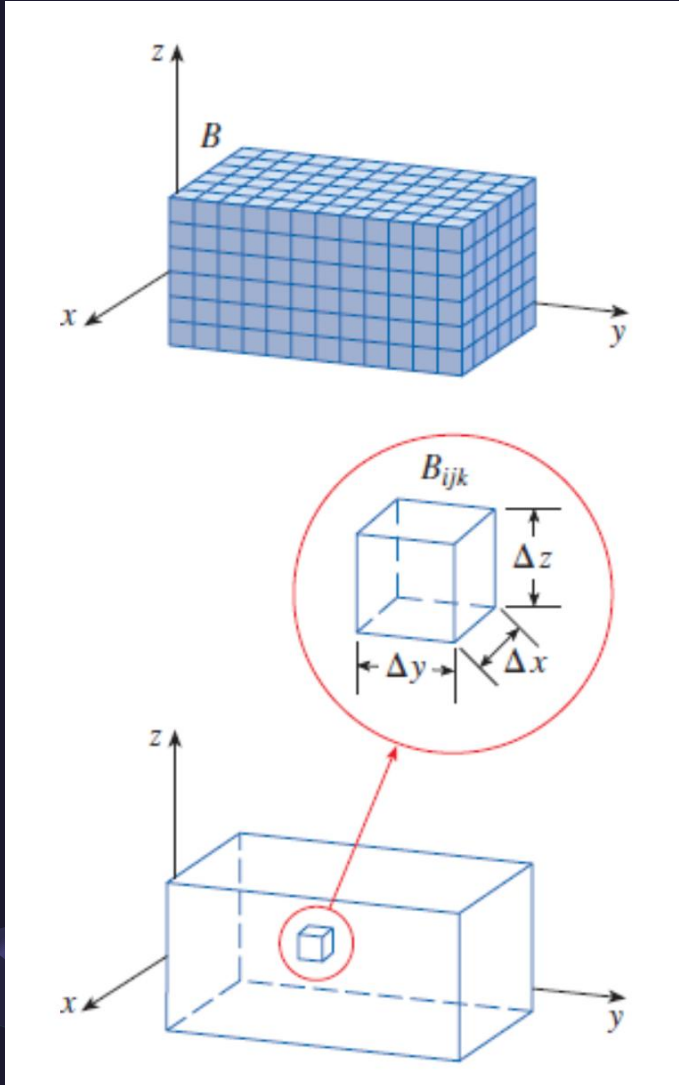
$$B = \{(x, y, z) \mid a \leq x \leq b, c \leq y \leq d, r \leq z \leq s\}$$

The **triple integral** of f over the box B is

$$\iiint_B f(x, y, z) \, dV = \lim_{l, m, n \rightarrow \infty} \sum_{i=1}^l \sum_{j=1}^m \sum_{k=1}^n f(x_{ijk}^*, y_{ijk}^*, z_{ijk}^*) \, \Delta V$$

if this limit exists.

Riemann Integral in 3 Variables



$$B = \{(x, y, z) \mid a \leq x \leq b, c \leq y \leq d, r \leq z \leq s\}$$

4 Fubini's Theorem for Triple Integrals

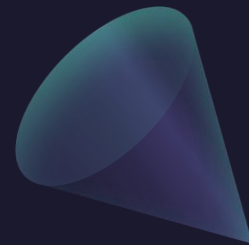
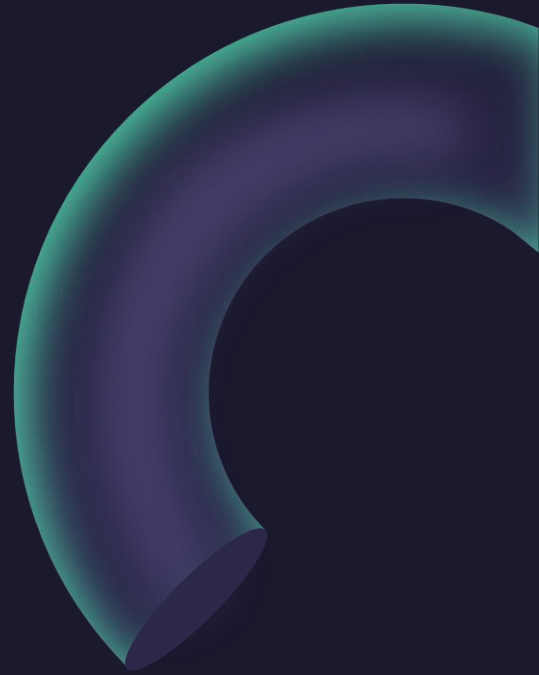
If f is continuous on the rectangular box $B = [a, b] \times [c, d] \times [r, s]$, then

$$\iiint_B f(x, y, z) \, dV = \int_r^s \int_c^d \int_a^b f(x, y, z) \, dx \, dy \, dz$$

Example: Evaluate the value of the triple integral

$$\iiint_B xyz^2 dV$$

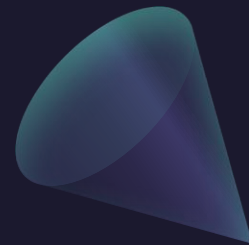
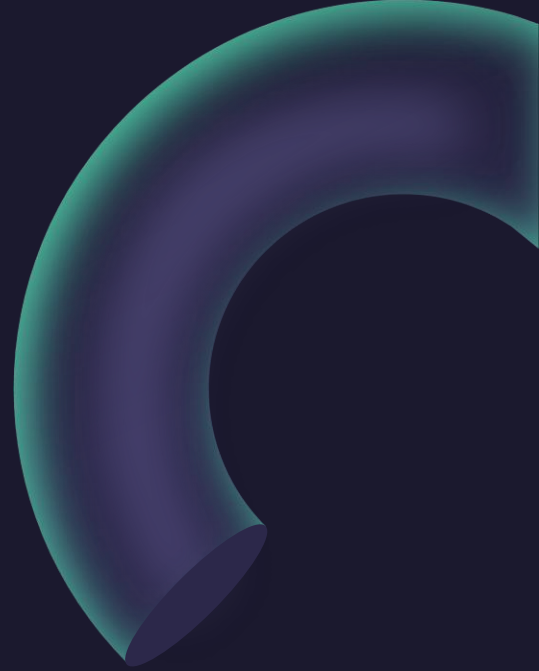
where B is the box $[0,1] \times [-1,2] \times [0,3]$.



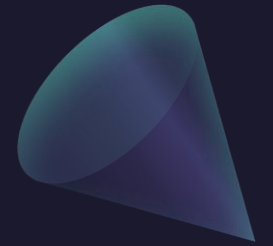
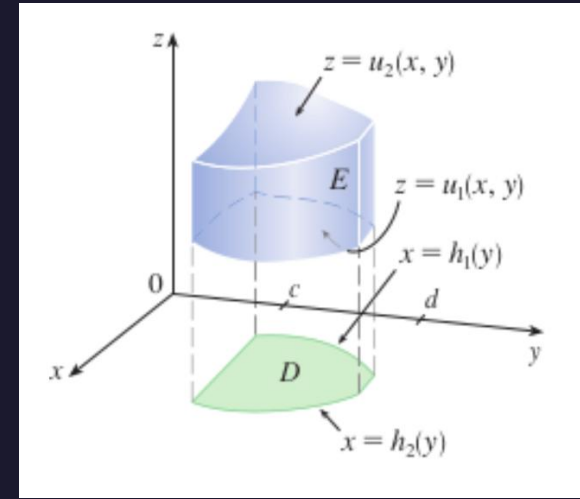
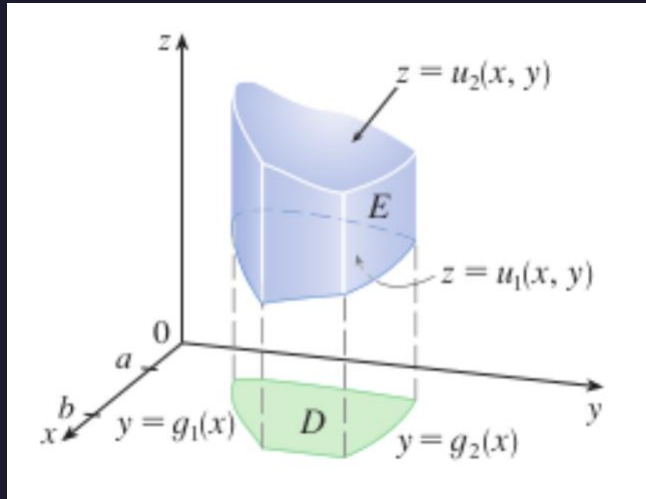
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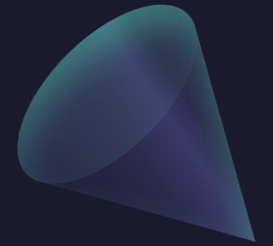
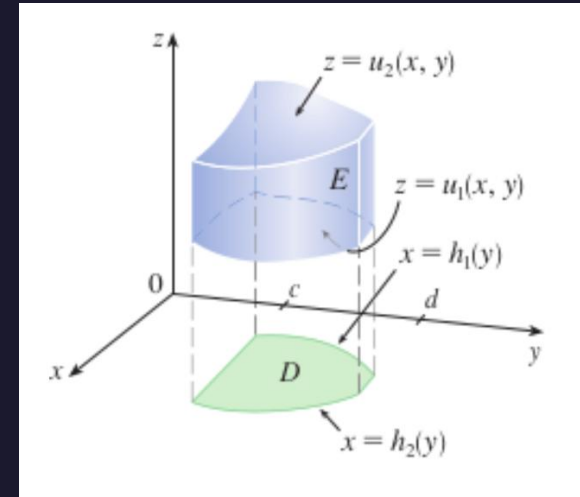
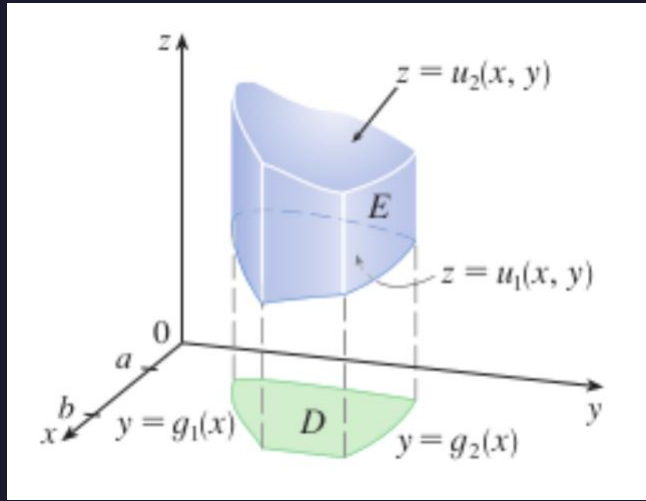
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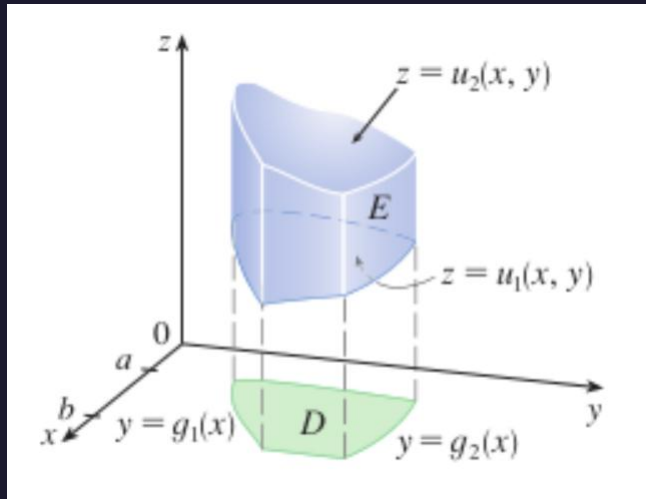
Triple Integrals in General Regions – Define the Region



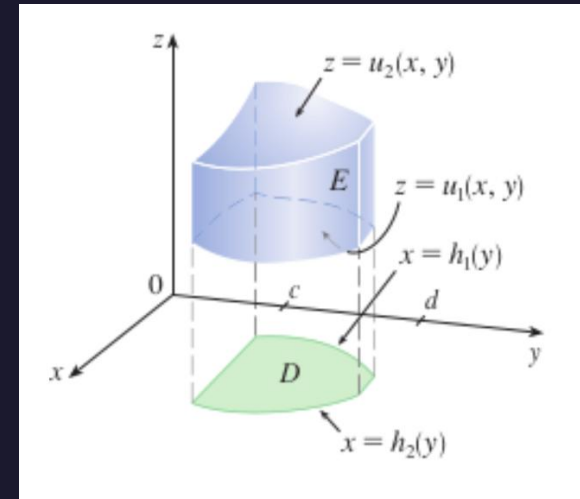
Triple Integrals in General Regions – Set up the Integral



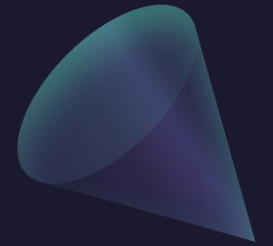
Triple Integrals in General Regions



$$\iiint_E f(x, y, z) \, dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \, dx \, dy$$



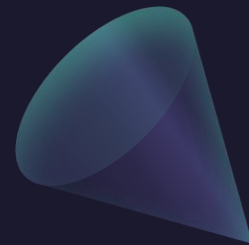
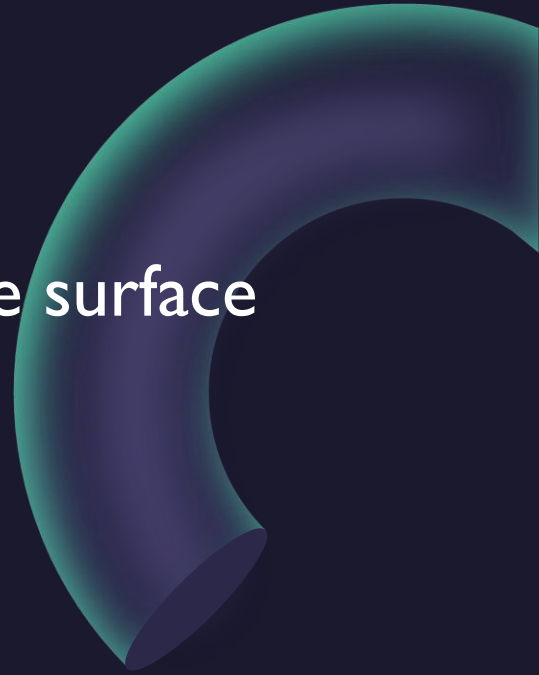
$$\iiint_E f(x, y, z) \, dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) \, dz \, dx \, dy$$



Example: Evaluate the value of the triple integral

$$\iiint_E z \, dV$$

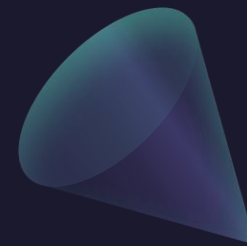
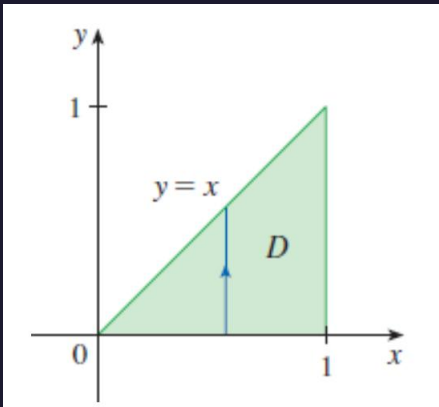
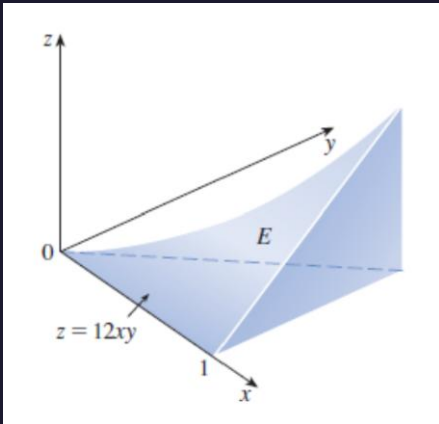
where E is the solid in the first octant bounded by the surface $z = 12xy$ and the planes $y = x$ and $x = 1$.



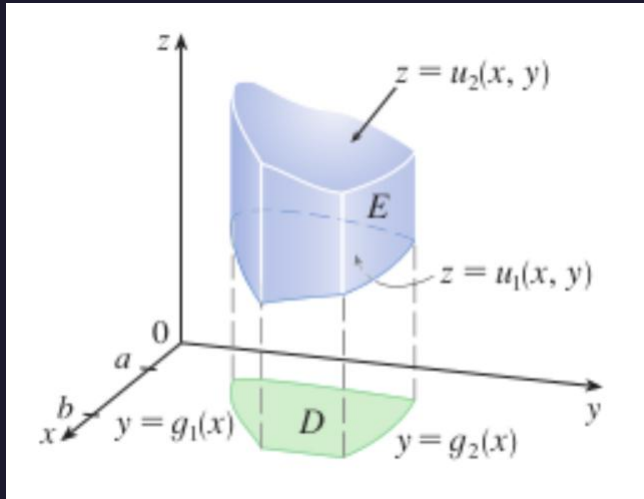
Example: Evaluate the value of the triple integral

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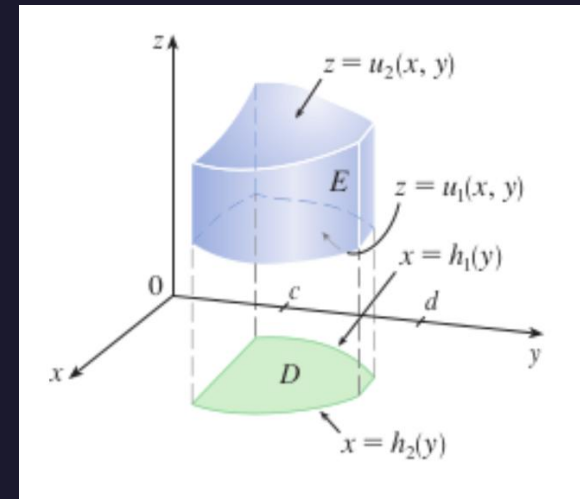
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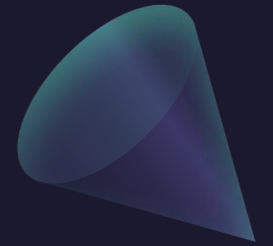
Triple Integrals in General Regions: Type I, ...



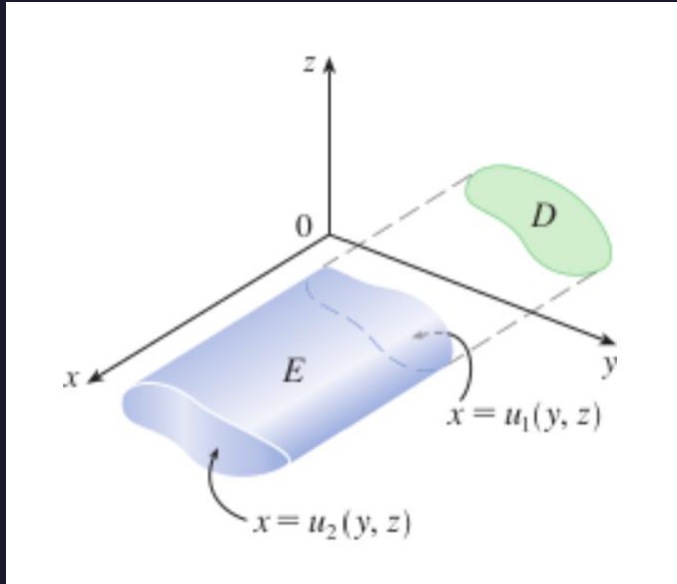
$$\iiint_E f(x, y, z) dV = \int_a^b \int_{g_1(x)}^{g_2(x)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz dx dy$$



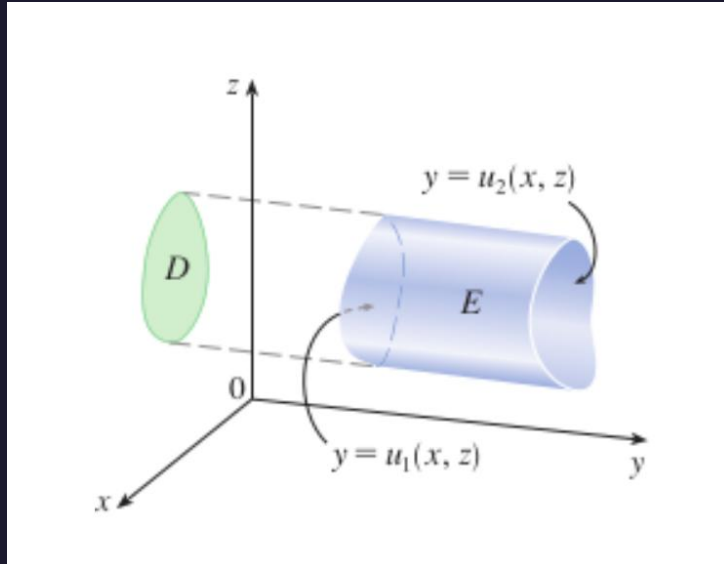
$$\iiint_E f(x, y, z) dV = \int_c^d \int_{h_1(y)}^{h_2(y)} \int_{u_1(x, y)}^{u_2(x, y)} f(x, y, z) dz dx dy$$



Triple Integrals in General Regions: Type II



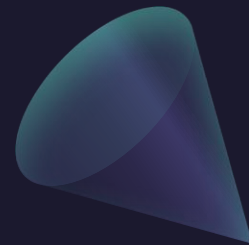
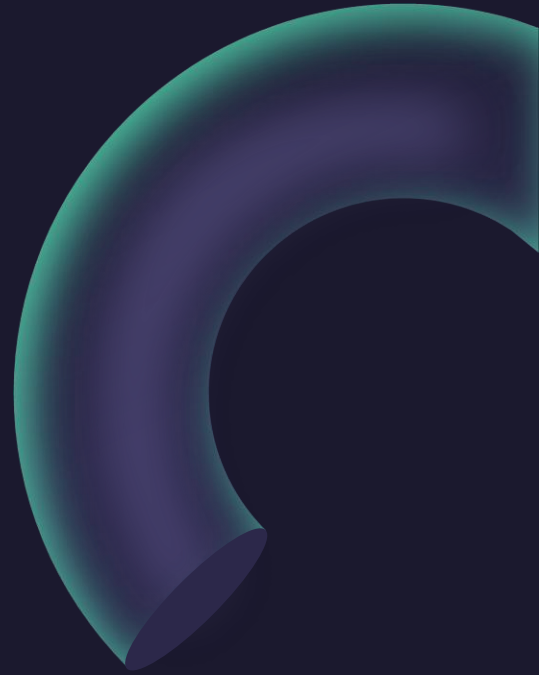
Triple Integrals in General Regions: Type III



Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + y^2} \, dV$$

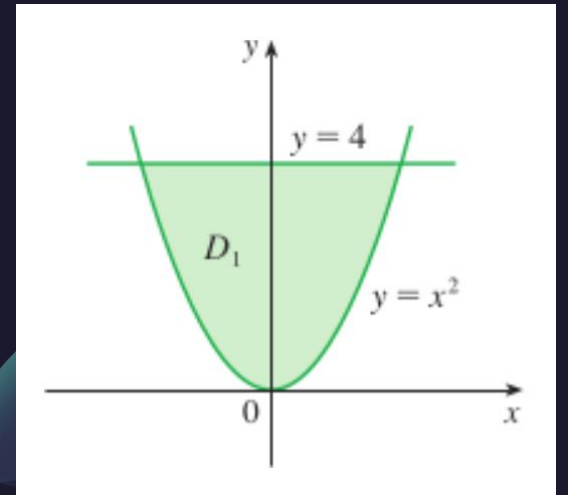
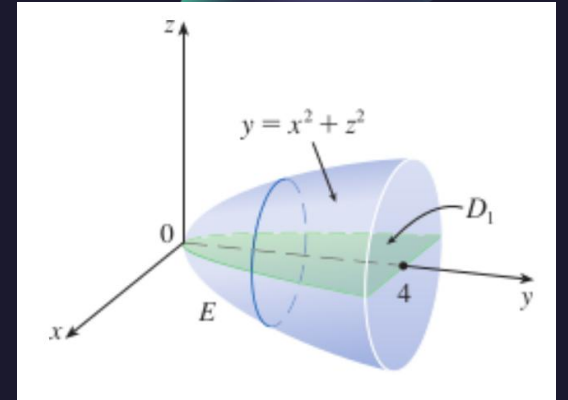
where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.



Example: Evaluate the value of the triple integral

$$\iiint_E \sqrt{x^2 + y^2} \, dV$$

where E is the solid bounded by the surface $y = x^2 + z^2$ and the plane $y = 4$.



Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

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Changing the Order of Integration

Example: Rewrite the following triple integral as a Type I, Type II, and Type III integrals

$$\int_0^1 \int_0^{x^2} \int_0^y (x + y + z) \, dz \, dy \, dx$$

Thank you

Until next time.

